Vincenzo Marcianò *Research Scientist*

marcianovincenzomv@gmail.com | SanBast | scholar.com/Ga_uQ98AAAAJ

Education

Visiting PhD Student, *King's College London* – London, UK — Sep 2023 - Jul 2026
PhD in AI & Computer Science, *Sorbonne University* – Biot, France — Sep 2023 - Jul 2026
MSc in Data Science & Engineering, *Polytechnic University of Turin* – Turin, Italy. GPA: 3.9/4.0 — Oct 2020 - Dec 2022
Exchange program for Msc Thesis, *The University of Sheffield* – Sheffield, United Kingdom — Feb 2022 - Jul 2022
BSc in Electronics Engineering, *Polytechnic University of Turin* – Turin, Italy. GPA: 3.0/4.0 — Oct 2017 - Jul 2020

Experience

Doctoral Researcher, *EURECOM (AI4Health Lab)* – Biot, France — Sep 2023 – Present

- Achieved a **15%** increase in segmentation quality correlation by developing *nnQC*, a multimodal CLIP-conditioned diffusion model enabling model-agnostic and efficient quality control.
- Improved segmentation accuracy by **10%** through implementing *HERMES*, a foundation-model-based framework leveraging optimal visual prompt sampling and embedded quality control.
- Contributed to **three open-source** medical imaging frameworks, for multimodal analysis.

Applied Scientist Intern, *Amazon* – Barcelona, Spain — Sep 2022 – Mar 2023

- Reduced forecasting error (WAPE) by **30%** by designing time-domain embeddings for multivariate time-series forecasting, enhancing accuracy and stability across vendor datasets.
- Improved model scalability and robustness by generalizing forecasting models across heterogeneous data granularities used in large-scale supply chain operations.

Research Scientist Intern, *Kroto Research Institute* – Sheffield, United Kingdom — May 2022 – Sep 2022

- Enhanced indoor–outdoor localization accuracy by **15%** by designing a lightweight ML system that extracts magnetometric time–frequency signal features for embedded applications.
- Identified significant gait–pathology correlations (Spearman's r = **0.42**) by engineering contextual time-series features to distinguish healthy and pathological motion patterns.

AI Engineer, *H2politO* – Turin, Italy — Oct 2021 – Aug 2022

- Enabled autonomous navigation for a hydrogen-powered car prototype by building a ROS-based NVIDIA-Jetson vision stack in C++, Python, and TensorFlow, leading to **4th place** at the Shell Eco Marathon.
- Achieved **2nd place** in the European Virtual Shell Eco Marathon by developing a self-driving virtual vehicle in the CARLA simulator for autonomous perception and control benchmarking.

Publications

- **Marcianò, V.**, et al. (2025) *Diffusion-Based Quality Control of Medical Image Segmentations across Different Organs*. Under review, IEEE TMI.
- Falcetta, D., **Marcianó, V.** et al. *VesselVerse: A Dataset and Collaborative Framework for Vessel Annotation.* In: Gee, J.C., et al. Medical Image Computing and Computer Assisted Intervention – MICCAI 2025.
- Poeta, E., Vargas, L., Falcetta, D., **Marcianó, V.** et al. (2025) *Divergence-Aware Training with Automatic Subgroup Mitigation for Breast Tumor Segmentation*. In: Gee, J.C., et al. MICCAI 2025 Workshops. Best Paper Award.
- Chaptoukaev, H., **Marcianó, V.**, Galati, F., Zuluaga, M.A. (2024). *HyperMM: Robust Multimodal Learning with Varying-Sized Inputs* MICCAI 2024 Workshops. Best Paper Award.
- **Marcianó V.** et al. (2024) *Discriminating Between Indoor and Outdoor Environments During Daily Living Activities Using Local Magnetic Field Characteristics and Machine Learning Techniques.* In: IEEE Sensors Journal.

Skills

Programming: Python (PyTorch, TorchIO, MONAI, huggingface, LangChain, scikit-learn, scikit-time, PySpark), C++, C
Tools: Docker, Git/GitHub, LaTeX, Linux/Unix, Bash, ITK Snap, HPC clusters, SQL, CMake, Make,
Generative Models: (Variational) AutoEncoders, GAN, Diffusion Models (DDPM, DDIM, Flow Matching), VLMs, LLMs.
Languages: Italian (Native), English (Advanced), French (Intermediate), Spanish (Basic)